

Engineering Orthogonal Alpha Through Structural Sterilization

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LinkedIn post snapshot

We stopped looking for "predictive" features we started looking for orthogonal ones our Sharpe doubled.

The raw data graveyards are full of brilliant-looking backtests that traded SPY with a 3.0 Sharpe and 50% correlation to the Momentum factor. That's not alpha; that's an expensive leveraged beta recreation.

In 2026, the edge isn't in finding a new price anomaly. The edge is in structural sterilization.

The advanced data workflow we use to actually get capacity allocated

1. The "Correlation to Nowhere" Test

Before we look at a signal's IC, we regress it against the "Big 5" macro regimes (Growth, Inflation, Risk Appetite, Liquidity, Tail Risk). If your novel shipping-container NLP signal has a 0.6 R-squared to the Manufacturing PMI surprise index, you don't have an alpha. You have a noisy proxy for the economic cycle. We reject candidates that can be explained by a 5-factor linear decomposition. Only pure residuals get to the backtest.

2. Time-Series Structure, Not Point-in-Time Price

Stop fitting to the EOD closing auction price. It's stale. We decompose the Limit Order Book into functional principal components (fPCA) to capture the shape of liquidity, not just the level. The mean is noise. The curvature of the book (gamma imbalance) is the signal. If your feature set is just OHLCV, you're training on the exhaust fumes of algorithms that are already smarter than you.

3. Gated Networks for Regime Qualification

We don't use one monolithic model. We use a Mixture of Experts with a dynamic gating network that reads volatility surfaces in real time. If the VIX futures term structure inverts (backwardation), the gating network instantly shifts the weighting from mean-reversion stat-arb features to momentum/trend features. The model selection is the alpha. Data scientists who train one stationary model and walk away are building ticking time bombs.

The goal is not to predict the future. The goal is to own risk that cannot be replicated by an ETF.

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